

A Ternary $(4k - 1(2))/3$ Collatz-Type Map

Complete 3-Adic Reduction, Algebraic Cycle Analysis, and Exhaustive Computational Verification up to 10^{11}

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Abstract

We study a ternary Collatz-type dynamical system on the positive integers not divisible by 3. In the compact notation of the title, $(4k - 1(2))/3$ means that the affine numerator subtracts 1 or 2 according to the residue class of the current state modulo 3; the actual accelerated map removes *all* factors of 3 at each step. More precisely, for

$$\mathcal{S} = \{x \in \mathbb{Z}_{>0} : 3 \nmid x\}, \quad r(x) = x \bmod 3 \in \{1, 2\},$$

we define

$$M(x) = 4x - r(x), \quad T(x) = \frac{M(x)}{3^{\nu_3(M(x))}}.$$

The map has the fixed points 1 and 2 and the positive four-cycle $22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22$. We derive exact residue branches and inverse branches, obtain a closed Diophantine equation for a general cycle, and show how the known positive and negative cycles are reconstructed algebraically. The four-cycle is governed by the near equality $4^4 = 256$ and $3^5 = 243$, giving the small cycle denominator $4^4 - 3^5 = 13$.

A geometric valuation heuristic gives $\Pr(\nu_3 = k) = 2/3^k$ for $k \geq 1$, hence $\mathbb{E}[\nu_3] = 3/2$ and the average logarithmic drift $\log 4 - \frac{3}{2} \log 3 \approx -0.261624$, with geometric factor about 0.7698. This is a probabilistic explanation for typical contraction, not a proof of global convergence.

Finally, we combine the previously archived exhaustive scan through 10^9 with a new disjoint V2 scan of $10^9 < x \leq 10^{11}$. The resulting verified domain contains exactly 66,666,666,667 admissible starts. The fixed point 1 attracts 6,602,925,203 starts (9.904388%), the fixed point 2 attracts 43,211,124,603 (64.816687%), and the four-cycle attracts 16,852,616,861 (25.278925%). No additional positive cycle, unresolved trajectory, or arithmetic overflow is reported in either component scan. Across the complete verified domain, the trajectories contain exactly 5,577,182,144,312 reduced-map iterations and remove 8,429,151,431,804 factors of 3 in total. The full-domain record transient is 574 reduced-map iterations from 97,312,923,676; the record accumulated 3-adic valuation is 744 from the same start; and the largest fully reduced state observed is 580,315,803,007,928,858,285, reached by a trajectory beginning at 52,991,345,843. The result is an exhaustive finite verification through 10^{11} , not a theorem for all positive integers.

Keywords: Collatz-type map; ternary dynamics; 3-adic valuation; accelerated iteration; Diophantine cycles; exhaustive computation.

1 Introduction

The classical Collatz problem is one of the simplest-looking and most persistent open problems in discrete mathematics. Its difficulty comes from the interaction between an affine growth rule

and residue-dependent division. Generalized Collatz systems replace parity by other moduli, use different affine coefficients, or compress several divisions into a single dynamical step. In these systems, arithmetic residue classes can determine both the typical drift and the possible periodic structures.

The present map is a ternary accelerated system. Every state is a positive integer not divisible by 3. The state is multiplied by 4, its residue modulo 3 is subtracted so that the result becomes divisible by 3, and then every available factor of 3 is removed. The complete removal of powers of 3 is the key structural feature: unlike an unaccelerated map that divides by only one factor of 3, one reduced iteration may remove several factors at once.

This distinction also creates several natural computational statistics. A trajectory has a number of reduced-map iterations, a total number of individual divisions by 3, and several possible notions of peak: the maximum fully reduced state, the maximum value after only the compulsory first division by 3, and the maximum raw affine value $M(x)$. These quantities should not be conflated.

The paper has four main aims. First, we formulate the map and its two residue branches precisely. Second, we analyze the valuation distribution and the associated logarithmic contraction heuristic. Third, we derive the exact cycle equation and reconstruct the known cycles algebraically. Fourth, building on the exhaustive 10^9 audit in the base article [3], we extend the verified positive domain to 10^{11} by combining that complete base interval with a disjoint V2 scan of $10^9 < x \leq 10^{11}$.

2 Definition of the domain and map

Let

$$\mathcal{S} = \{x \in \mathbb{Z}_{>0} : x \not\equiv 0 \pmod{3}\}. \quad (1)$$

Every $x \in \mathcal{S}$ has exactly one residue

$$r(x) \in \{1, 2\}, \quad r(x) \equiv x \pmod{3}.$$

Define the affine pre-reduction map

$$M(x) = 4x - r(x) = \begin{cases} 4x - 1, & x \equiv 1 \pmod{3}, \\ 4x - 2, & x \equiv 2 \pmod{3}. \end{cases} \quad (2)$$

Since $4 \equiv 1 \pmod{3}$, we have $4x - r(x) \equiv 0 \pmod{3}$. Therefore $M(x)$ always contains at least one factor of 3.

For $n \neq 0$, let

$$\nu_3(n) = \max\{j \geq 0 : 3^j \mid n\}$$

be the 3-adic valuation. The accelerated map is

$$T(x) = \frac{4x - r(x)}{3^{\nu_3(4x - r(x))}} \quad (3)$$

and is well-defined from $\mathcal{S}$ to itself.

2.1 The title notation $(4k - 1(2))/3$

The compact notation in the title records the two ternary affine branches: subtract 1 when the state is $1 \pmod{3}$ and subtract 2 when it is $2 \pmod{3}$. Equation (3) is the precise definition used throughout the mathematics and computation. The denominator 3 in the compact title emphasizes the compulsory ternary division, while the actual reduced map divides by the full power 3^{ν_3} .

3 Algebra of the two residue branches

The branch structure becomes especially transparent after writing the state in residue form.

If $x = 3q + 1$, then

$$M(x) = 4(3q + 1) - 1 = 12q + 3 = 3(4q + 1), \quad (4)$$

so

$$T(3q + 1) = \frac{4q + 1}{3^{\nu_3(4q+1)}}. \quad (5)$$

Likewise, if $x = 3q + 2$, then

$$M(x) = 4(3q + 2) - 2 = 12q + 6 = 3(4q + 2), \quad (6)$$

and therefore

$$T(3q + 2) = \frac{4q + 2}{3^{\nu_3(4q+2)}}. \quad (7)$$

Equations (5)–(7) show explicitly that one factor of 3 is compulsory in the affine form, while any additional factors depend on the smaller branch quantities $4q + 1$ and $4q + 2$.

3.1 Exact inverse branches

The one-step relation can be written

$$3^k y = 4x - r, \quad r \in \{1, 2\}, \quad k \geq 1, \quad (8)$$

where $y = T(x)$. Hence every possible predecessor associated with a prescribed residue r and valuation k has the form

$$x = \frac{3^k y + r}{4}. \quad (9)$$

Such a predecessor exists exactly when the numerator is divisible by 4 and its valuation is exactly compatible with the chosen k . Because $4 \equiv 1 \pmod{3}$, a valid integer in (9) automatically has residue r modulo 3. This inverse-branch representation is useful both for cycle algebra and for constructing the preimage tree of an attractor.

4 Step-by-step numerical examples

The smallest states already reveal all three positive attractors.

For $x = 1$,

$$M(1) = 4 - 1 = 3, \quad \nu_3(3) = 1, \quad T(1) = 1,$$

so 1 is a fixed point. For $x = 2$,

$$M(2) = 8 - 2 = 6, \quad \nu_3(6) = 1, \quad T(2) = 2,$$

so 2 is also a fixed point.

For $x = 4$,

$$M(4) = 16 - 1 = 15 = 3 \cdot 5, \quad T(4) = 5.$$

For $x = 5$,

$$M(5) = 20 - 2 = 18 = 3^2 \cdot 2, \quad T(5) = 2,$$

so $4 \rightarrow 5 \rightarrow 2$. For $x = 7$,

$$M(7) = 28 - 1 = 27 = 3^3, \quad T(7) = 1,$$

showing how a high valuation can produce a sharp fall in a single reduced step.

The nontrivial positive cycle is

$$22 \longrightarrow 29 \longrightarrow 38 \longrightarrow 50 \longrightarrow 22. \quad (10)$$

Indeed,

$$\begin{aligned} 4(22) - 1 &= 87 = 3 \cdot 29, \\ 4(29) - 2 &= 114 = 3 \cdot 38, \\ 4(38) - 2 &= 150 = 3 \cdot 50, \\ 4(50) - 2 &= 198 = 3^2 \cdot 22. \end{aligned}$$

Thus three steps remove one factor of 3, while the final step removes two. The valuation vector is $(1, 1, 1, 2)$ and its sum is $K = 5$.

5 The 3-adic valuation heuristic

The large-scale behavior is strongly influenced by the number of powers of 3 removed in each reduced iteration. Suppose, heuristically, that after the compulsory divisibility by 3, the remaining residue information is distributed uniformly modulo higher powers of 3. Then for $k \geq 1$,

$$\Pr(\nu_3(M(x)) = k) = \left(\frac{1}{3}\right)^{k-1} \left(1 - \frac{1}{3}\right) = \frac{2}{3^k}. \quad (11)$$

This is a geometric distribution on $\{1, 2, 3, \dots\}$.

5.1 Expected valuation

From (11),

$$\mathbb{E}[\nu_3] = \sum_{k=1}^{\infty} k \frac{2}{3^k} = \frac{3}{2}. \quad (12)$$

Thus the heuristic average denominator in multiplicative scale is $3^{3/2} = 3\sqrt{3} \approx 5.196$, while the numerator has leading coefficient 4.

5.2 Average logarithmic drift

For large x , the subtraction of $r(x) \in \{1, 2\}$ is lower order, so

$$\log \frac{T(x)}{x} \approx \log 4 - \nu_3 \log 3.$$

Taking the heuristic expectation gives

$$\Delta_{\text{heur}} = \log 4 - \frac{3}{2} \log 3 \approx -0.261624. \quad (13)$$

The associated geometric factor is

$$\lambda_{\text{heur}} = e^{\Delta_{\text{heur}}} = \frac{4}{3^{3/2}} = \frac{4}{3\sqrt{3}} \approx 0.769800. \quad (14)$$

In this model a typical logarithmic step contracts the scale by about 23%.

5.3 What the heuristic does not prove

The negative drift gives a useful statistical explanation for the computational behavior, but it is not a proof that every deterministic orbit converges. Successive valuations along a fixed orbit need not behave as independent samples from (11), and a negative mean drift does not by itself exclude exceptional unbounded trajectories. The exhaustive results below are therefore stated only for the finite domain actually tested.

6 Algebraic cycle equation

This section gives the exact arithmetic structure behind every periodic orbit of the map.

Suppose a cycle has length L and states

$$x_0, x_1, \dots, x_{L-1}, \quad x_L = x_0.$$

At step i , define

$$r_i = x_i \bmod 3 \in \{1, 2\}, \quad k_i = \nu_3(4x_i - r_i) \geq 1.$$

Then

$$3^{k_i} x_{i+1} = 4x_i - r_i. \tag{15}$$

Let

$$K = \sum_{i=0}^{L-1} k_i. \tag{16}$$

Repeated substitution of (15) around the entire loop gives

$$x_0 = \frac{\sum_{j=0}^{L-1} r_j 4^{L-1-j} 3^{\sum_{m=0}^{j-1} k_m}}{4^L - 3^K} \tag{17}$$

with the empty exponent sum interpreted as 0.

Equation (17) gives a necessary exact divisibility condition: for a prescribed residue/valuation word, $4^L - 3^K$ must divide the numerator, and the resulting integer states must reproduce the assumed residues and valuations.

6.1 Sign condition and the critical ratio

The numerator of (17) is positive. Hence a positive cycle requires

$$4^L - 3^K > 0 \iff \frac{K}{L} < \frac{\log 4}{\log 3} \approx 1.261860. \tag{18}$$

A negative cycle instead requires

$$4^L - 3^K < 0 \iff \frac{K}{L} > \frac{\log 4}{\log 3}. \tag{19}$$

Thus the rational approximation of $\log 4 / \log 3$ by K/L is built directly into the existence problem for cycles.

7 Algebraic reconstruction of the known cycles

7.1 The fixed points

For a one-cycle, $L = 1$ and $K = 1$. With $r_0 = 1$,

$$x_0 = \frac{1}{4-3} = 1.$$

With $r_0 = 2$,

$$x_0 = \frac{2}{4-3} = 2.$$

Thus the two positive fixed points arise immediately from the cycle equation.

7.2 The positive four-cycle

For (10),

$$L = 4, \quad (r_0, r_1, r_2, r_3) = (1, 2, 2, 2), \quad (k_0, k_1, k_2, k_3) = (1, 1, 1, 2),$$

so $K = 5$. The denominator is

$$4^4 - 3^5 = 256 - 243 = 13. \quad (20)$$

The numerator is

$$1 \cdot 4^3 + 2 \cdot 4^2 \cdot 3 + 2 \cdot 4 \cdot 3^2 + 2 \cdot 3^3 = 64 + 96 + 72 + 54 = 286.$$

Hence

$$x_0 = \frac{286}{13} = 22, \quad (21)$$

and direct iteration reconstructs $22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22$.

The structural point is the unusually small denominator produced by

$$4^4 = 256 \approx 243 = 3^5. \quad (22)$$

Equivalently, $K/L = 5/4 = 1.25$ is close to the critical ratio $\log 4 / \log 3 \approx 1.26186$. This Diophantine near-balance makes the four-cycle arithmetically possible.

7.3 Negative cycles as an algebraic extension

Although the exhaustive 10^{11} audit in this paper concerns positive starting values, the same cycle equation also reconstructs the negative cycles noted in the earlier analysis.

For the two-cycle $-1 \rightarrow -2 \rightarrow -1$, starting at -1 gives

$$(r_0, r_1) = (2, 1), \quad (k_0, k_1) = (1, 2), \quad L = 2, \quad K = 3.$$

Then

$$4^2 - 3^3 = 16 - 27 = -11, \quad 2 \cdot 4 + 1 \cdot 3 = 11,$$

and therefore $x_0 = 11/(-11) = -1$.

For the seven-cycle

$$-7 \rightarrow -10 \rightarrow -14 \rightarrow -19 \rightarrow -26 \rightarrow -35 \rightarrow -47 \rightarrow -7, \quad (23)$$

we have

$$L = 7, \quad K = 9,$$

with residue vector $(2, 2, 1, 2, 1, 1, 1)$ and valuation vector $(1, 1, 1, 1, 1, 1, 3)$. The denominator is

$$4^7 - 3^9 = 16384 - 19683 = -3299,$$

while the numerator in (17) is 23093. Thus

$$x_0 = \frac{23093}{-3299} = -7,$$

which reconstructs the full seven-cycle exactly. These negative examples emphasize that the sign of $4^L - 3^K$ cleanly separates the arithmetic regimes of positive and negative cycles.

8 Computational methodology

The finite verification through 10^{11} is the union of two disjoint exhaustive computations. The first component is the previously archived scan from the base article [3], covering

$$1 \leq x \leq 10^9, \quad 3 \nmid x,$$

with exactly 666,666,667 admissible starts. The new V2 extension covers

$$10^9 < x \leq 10^{11}, \quad 3 \nmid x,$$

with exactly 66,000,000,000 admissible starts. Their union is therefore the complete finite domain

$$1 \leq x \leq 10^{11}, \quad 3 \nmid x,$$

whose exact cardinality is

$$10^{11} - \left\lfloor \frac{10^{11}}{3} \right\rfloor = 66,666,666,667. \quad (24)$$

The V2 extension scanner is implemented in C and iterates the exact accelerated map (3). Orbit arithmetic uses `unsigned __int128`; a 64-bit fast path is used only when the affine transition is provably safe in `uint64_t`, with automatic promotion to 128-bit arithmetic otherwise. The scanner checks the two fixed points and the known four-cycle before each transition, and a repeated state outside those known attractors is detected by an exact constant-memory cycle detector and recorded as an unknown cycle. There is no iteration cap in the V2 production loop: a trajectory terminates only at a known attractor, a newly detected cycle, or an arithmetic boundary event.

The extension interval was divided into 990 raw-integer chunks of size 100,000,000 and scanned with four POSIX worker threads. Each completed chunk is written to a `.done` checkpoint by atomic rename. Re-running the scanner skips completed chunks and reconstructs the session totals from the saved records. Every checkpoint stores the attractor counts, unknown-cycle and overflow counts, total reduced steps, total removed 3-adic valuation, and the chunk maxima for transient length, accumulated valuation, and fully reduced peak.

9 V2 validation and reproducibility checks

Before the 10^{11} extension, the V2 implementation was run on

$$1 \leq x \leq 10^6,$$

using ten raw-integer chunks of size 100,000. It exactly reproduces the validation statistics in the base article: 666,667 admissible starts, with the basin counts shown below.

Attractor	Count	Percentage
$\{1\}$	66,382	9.957295%
$\{2\}$	431,888	64.783168%
$22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22$	168,397	25.259537%

The same validation recovers the 249-iteration transient record from 969,812, the accumulated valuation record 326 from the same start, and the fully reduced peak 68,682,968,834 from 491,732.

A separate V2 benchmark run on the first production chunk

$$1,000,000,001 \leq x \leq 1,100,000,000$$

reproduced the saved production checkpoint for that interval in all integer-valued statistics. This duplicate calculation provides an additional check on the production executable and checkpoint format.

10 New extension scan: $10^9 < x \leq 10^{11}$

The 990 production checkpoints cover the extension interval contiguously from 1,000,000,001 through 100,000,000,000. Summing their saved statistics gives

$$\boxed{\text{tested} = 66,000,000,000, \quad \text{unknown} = 0, \quad \text{overflow} = 0.} \quad (25)$$

No terminal positive cycle other than the two fixed points and the four-cycle was found.

Table 1: Attractor classification in the new interval $10^9 < x \leq 10^{11}$, $3 \nmid x$.

Attractor	Number of starts	Percentage
$\{1\}$	6,536,905,587	9.904402%
$\{2\}$	42,779,024,148	64.816703%
$22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22$	16,684,070,265	25.278894%
Total	66,000,000,000	100.000000%

The exact count identity

$$6,536,905,587 + 42,779,024,148 + 16,684,070,265 = 66,000,000,000$$

accounts for every admissible start in the extension interval. The same checkpoints record a total of 5,533,148,453,392 reduced-map iterations and a total accumulated 3-adic valuation of 8,362,466,366,261 over the extension scan.

The extension produces new finite-domain records. The longest reduced-map transient is

$$\boxed{574 \text{ iterations from } x_0 = 97,312,923,676,} \quad (26)$$

and the accumulated number of removed factors of 3 from the same start is

$$\boxed{744.} \quad (27)$$

The largest fully reduced state recorded in the extension is

$$\boxed{580,315,803,007,928,858,285,} \quad (28)$$

reached on the trajectory beginning at

$$\boxed{x_0 = 52,991,345,843.} \quad (29)$$

All three records exceed the corresponding records in the base 10^9 computation.

11 Combined exhaustive verification through 10^{11}

The base interval and the new extension are disjoint and together exhaust every positive admissible start through 10^{11} . Adding the base counts [3] to Table 1 gives the complete classification.

Table 2: Complete attractor classification for all $1 \leq x \leq 10^{11}$ with $3 \nmid x$.

Attractor	Number of starts	Percentage
$\{1\}$	6,602,925,203	9.904388%
$\{2\}$	43,211,124,603	64.816687%
$22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22$	16,852,616,861	25.278925%
Total	66,666,666,667	100.000000%

The combined count identity

$$6,602,925,203 + 43,211,124,603 + 16,852,616,861 = 66,666,666,667$$

is exact. Since both component scans report zero unknown cycles and zero arithmetic overflows, the same is true of the combined finite verification. Every tested start through 10^{11} enters one of the two fixed points or the positive four-cycle.

The V2 scanner was also applied to the complete base interval $1 \leq x \leq 10^9$ as a supplementary aggregate verification. It reproduces the archived base basin counts and all three published base extrema exactly, while additionally recording

$$\text{base total reduced steps} = 44,033,690,920, \quad \text{base total } \nu_3 = 66,685,065,543.$$

The base total-step value independently agrees with the exact integer recovered from the archived chunk means in the original 10^9 computational package. Adding these base aggregates to the disjoint V2 extension totals gives the following full-domain statistics.

Table 3: Aggregate trajectory statistics for the two disjoint computational intervals and their union.

Quantity	Base $x \leq 10^9$	Extension	Combined $x \leq 10^{11}$
Admissible starts	666,666,667	66,000,000,000	66,666,666,667
Total reduced iterations	44,033,690,920	5,533,148,453,392	5,577,182,144,312
Mean reduced iterations/start	66.050536347	83.835582627	83.657732164
Total removed 3-adic factors	66,685,065,543	8,362,466,366,261	8,429,151,431,804
Mean removed factors/start	100.027598264	126.704035852	126.437271476
Unknown cycles	0	0	0
Arithmetic overflows	0	0	0

For transparency, the arithmetic decomposition of the three basin counts is

Attractor	Base $x \leq 10^9$	Extension	Combined
$\{1\}$	66,019,616	6,536,905,587	6,602,925,203
$\{2\}$	432,100,455	42,779,024,148	43,211,124,603
Four-cycle	168,546,596	16,684,070,265	16,852,616,861
Total	666,666,667	66,000,000,000	66,666,666,667

12 Comparison across verified scales

The basin proportions remain numerically stable as the verified domain expands from 10^6 to 10^9 and then to 10^{11} . The proportions move only slightly across these ranges. This is compatible

Table 4: Observed positive basin proportions across three exhaustive cutoffs.

Attractor	$x \leq 10^6$	$x \leq 10^9$	$x \leq 10^{11}$
$\{1\}$	9.957295%	9.902942%	9.904388%
$\{2\}$	64.783168%	64.815068%	64.816687%
Four-cycle	25.259537%	25.281989%	25.278925%

with the existence of limiting basin densities, but the finite experiments neither prove that such limits exist nor determine exact limiting values.

The extremal orbit statistics, by contrast, continue to grow sharply. Thus stable aggregate

Table 5: Growth of selected finite-domain records.

Cutoff	Max. reduced steps	Max. accumulated ν_3	Max. reduced peak
10^6	249	326	68,682,968,834
10^9	405	529	164,210,561,845,359,809
10^{11}	574	744	580,315,803,007,928,858,285

basin proportions coexist with increasingly extreme individual excursions.

13 Interpretation of the cycle structure

Three mechanisms interact in the dynamics.

First, the valuation distribution favors contraction on the logarithmic average scale. Second, the exact cycle equation imposes a strong divisibility constraint on every periodic residue-valuation word. Third, the balance between 4^L and 3^K controls the size and sign of the cycle denominator. The positive four-cycle is a particularly transparent example because $4^4 - 3^5 = 13$ is unusually small.

This suggests a useful search principle for periodic orbits: rational approximations K/L near $\log 4 / \log 3$ produce small values of $|4^L - 3^K|$ and hence arithmetically favorable candidates, but a candidate still has to satisfy the exact numerator divisibility and reproduce its residue and valuation pattern. The near equality alone is therefore not sufficient for a cycle.

14 Finite theorem and global conjecture

The combined computation establishes the following finite statement.

Finite computational statement. For every $x \in \mathcal{S}$ with $1 \leq x \leq 10^{11}$, the exact finite audit represented by the base computation [3] together with the archived V2 extension reaches one of

$$\{1\}, \quad \{2\}, \quad 22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22,$$

and neither component computation reports an arithmetic overflow or an additional terminal cycle.

The numerical evidence motivates the same global conjecture as before.

Conjecture (positive three-attractor convergence). Every positive integer not divisible by 3, under the map (3), eventually enters one of the two fixed points 1, 2 or the four-cycle $22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22$.

The exhaustive finite verification through 10^{11} is substantially stronger evidence than the earlier 10^9 audit, but it does not prove the conjecture. A proof must exclude every additional positive cycle at all larger scales and rule out unbounded trajectories.

15 What remains for a proof?

A complete proof would still need to overcome at least three distinct obstacles.

1. The geometric distribution (11) is a probabilistic model. A deterministic proof needs sufficient control of the actual valuation frequencies along every orbit, or an argument that does not depend on statistical independence.
2. The cycle equation (17) must be controlled for arbitrarily large L and K . Finite searches over lengths cannot exclude a cycle with much larger parameters.
3. Large temporary excursions must be handled uniformly. The observed reduced peak above 5.8×10^{20} from a start below 10^{11} demonstrates that early descent or small local growth cannot serve as a global argument.

These limitations separate the rigorous finite computation from the still-open all-integers statement.

16 Reproducibility

The computational archive accompanying this manuscript contains only V2 materials: the V2 C source and corresponding executable, ten validation checkpoints, one benchmark checkpoint, all 990 production checkpoints for the $10^9 < x \leq 10^{11}$ extension, ten supplementary V2 checkpoints covering the base interval $1 \leq x \leq 10^9$, a combined aggregate summary, and the final LaTeX source of this manuscript. The PDF is distributed separately. No V1, V3, V4, or V5 scanner files are included.

A typical V2 compilation command is

```
clang -O3 -march=native -pthread scan_ternary_minus_1e11_v2.c \
-o scan_ternary_minus_1e11_v2
```

The production extension scan is represented by

```
./scan_ternary_minus_1e11_v2 1000000001 100000000000 \
1000000000 4 results_1e11_chunks
```

and the validation run through 10^6 by

```
./scan_ternary_minus_1e11_v2 1 1000000 100000 4 \
test_v2_chunks
```

The supplementary V2 base verification is reproduced by

```
./scan_ternary_minus_1e11_v2 1 1000000000 100000000 4 \
base_1e9_v2_verification
```

The archived chunk files themselves are the machine-readable audit trail. Independent aggregation of the 990 production records should recover the extension counts in Table 1, zero unknown cycles, zero overflow events, the 574-iteration transient record, the accumulated valuation record 744, and the maximum fully reduced state reported above. Aggregating the ten supplementary base checkpoints reproduces the published 10^9 basin counts and extrema and yields the base totals in Table 3; summing the two disjoint intervals yields the combined totals 5,577,182,144,312 reduced iterations and 8,429,151,431,804 removed factors of 3.

17 Conclusion

The ternary $(4k - 1(2))/3$ Collatz-type system studied here is most precisely represented by the accelerated map

$$T(x) = \frac{4x - (x \bmod 3)}{3^{\nu_3(4x - (x \bmod 3))}}, \quad 3 \nmid x.$$

Its local rule is elementary, but the resulting arithmetic dynamics combine residue branching, variable 3-adic compression, large transient excursions, and nontrivial periodic structure.

The theoretical analysis is unchanged in character from the base study: the geometric valuation model gives a negative average logarithmic drift and the exact cycle equation converts periodicity into a Diophantine divisibility problem controlled by the balance of 4^L and 3^K . The new contribution is computational scale. Combining the archived 10^9 audit with the disjoint V2 extension classifies all 66,666,666,667 admissible positive starts through 10^{11} . Across this complete domain the trajectories contain 5,577,182,144,312 reduced iterations and 8,429,151,431,804 removed factors of 3. No additional positive attractor, unresolved cycle, or overflow is reported, while the record transient, accumulated valuation, and reduced peak all increase substantially. The resulting three-attractor convergence conjecture nevertheless remains open beyond the finite verified range.

Data and code availability

The Zenodo deposit for this 10^{11} study is intended to contain three deliverables: the PDF manuscript, the standalone LaTeX source, and a V2-only computational archive containing the scanner, validation and benchmark checkpoints, the complete 990-chunk extension audit trail, the ten-checkpoint supplementary V2 base verification, the combined aggregate summary, and the same LaTeX source. The earlier 10^9 paper and its finite results remain independently citable through its Zenodo DOI below.

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